

Total no. of Pages: 4

Roll no. ....

Degree: B. Tech

Semester: 3<sup>rd</sup>

END-SEMESTER THEORY EXAMINATION, NOV-DEC, 2024

Course Code: ITECC305/INECC305/ITECC06/INECC06

Course Title: Probability Theory and Stochastic Process

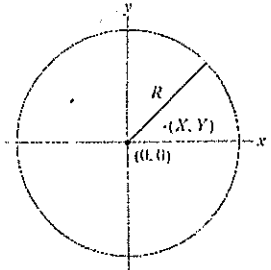
Time: 03 Hours

Max. Marks: 50

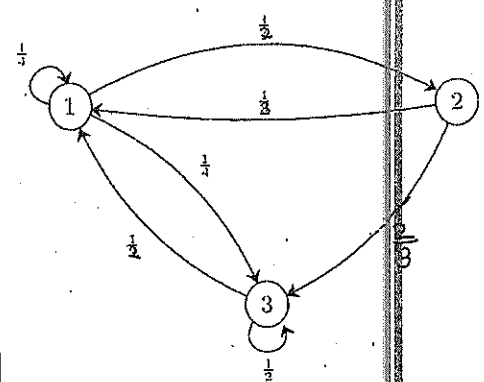
Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

| Q. No. | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks | CO  |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|
| Q 1    | Attempt any 2 parts of the following                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 10    |     |
| 1a     | A bin contains 3 different types of disposable flashlights. The probability that a type 1 flashlight will give over 100 hours of use is .7, with the corresponding probabilities for type 2 and type 3 flashlights being .4 and .3, respectively. Suppose that 20 percent of the flashlights in the bin are type 1, 30 percent are type 2, and 50 percent are type 3.<br>(i) What is the probability that a randomly chosen flashlight will give more than 100 hours of use?<br>(ii) Given that a flashlight lasted over 100 hours, what is the conditional probability that it was a type $j$ flashlight, $j = 1, 2, 3$ ?                                                                                                                                                                                                         | 5     | CO1 |
| 1b     | (i) At a certain stage of a criminal investigation, the inspector in charge is 60 percent convinced of the guilt of a certain suspect. Suppose, however, that a new piece of evidence which shows that the criminal has a certain characteristic (such as left-handedness, baldness, or brown hair) is uncovered. If 20 percent of the population possesses this characteristic, how certain of the guilt of the suspect should the inspector now be if it turns out that the suspect has the characteristic?<br>(ii) A system consisting of $n$ separate components is said to be a series system if it functions when all $n$ components function. Assume that the components fail independently and that the probability of failure of component $k$ is $p_k$ , $k=1,2,...,n$ . Find the probability that the system functions. | 5     | CO1 |
| 1c     | (i) An urn contains two type $A$ coins and one type $B$ coin. When a type $A$ coin is flipped, it comes up heads with probability $1/4$ , whereas when a type $B$ coin is flipped, it comes up heads with probability $3/4$ . A coin is randomly chosen from the urn and flipped. Given that the flip landed on heads, what is the probability that it was a type $A$ coin?<br>(ii) It is known that screws produced by a certain company will be defective with probability .01, independently of each other. The company sells the screws in packages of 10 and offers a money-back guarantee that at most 1 of the 10 screws is defective. What proportion of packages sold must the company replace?                                                                                                                           | 5     | CO1 |
| Q 2    | Attempt any 2 parts of the following                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 10    |     |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |     |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|
| 2a  | (i) The probability mass function of a random variable $X$ is given by $p(i) = c\lambda^i/i!$ , $i = 0, 1, 2, \dots$ , where $\lambda$ is some positive value. Find $P\{X=0\}$ and $P\{X>2\}$ .<br>(ii) A school class of 120 students is driven in 3 buses to a symphonic performance. There are 36 students in one of the buses, 40 in another, and 44 in the third bus. When the buses arrive, one of the 120 students is randomly chosen. Let $X$ denote the number of students on the bus of that randomly chosen student, and find $E[X]$ (till 2nd decimal place).                                                                                                                | 5  | CO2 |
| 2b  | (i) Consider an experiment that consists of counting the number of $\alpha$ particles given off in a 1-second interval by 2 gram of radioactive material. If we know from past experience that, on the average, 3.2 such $\alpha$ particles are given off, what is a good approximation to the probability that no more than 2 $\alpha$ particles will appear?<br>(ii) Consider a function<br>$f(x) = \frac{1}{\sqrt{\pi}} e^{(-x^2+x-a)}$ , $-\infty < x < \infty$<br>Find the value of 'a' such that $f(x)$ is a pdf of a continuous r.v. $X$                                                                                                                                          | 5  | CO2 |
| 2c  | For each of the following random variables, find the Moment Generating Function.<br>(i) $X$ is a discrete random variable, with PMF<br>$P_X(k) = \begin{cases} \frac{1}{3}, & k = 1 \\ \frac{2}{3}, & k = 2 \end{cases}$<br>(ii) $Y$ is a Uniform (0,1) random variable.                                                                                                                                                                                                                                                                                                                                                                                                                 | 5  | CO2 |
| Q 3 | Attempt any 2 parts of the following                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 10 |     |
| 3a  | Consider a circle of radius $R$ , and suppose that a point within the circle is randomly chosen in such a manner that all regions within the circle of equal area are equally likely to contain the point. (In other words, the point is uniformly distributed within the circle.) If we let the center of the circle denote the origin and define $X$ and $Y$ to be the coordinates of the point chosen (Figure below), then, since $(X, Y)$ is equally likely to be near each point in the circle, it follows that the joint density function of $X$ and $Y$ is given by<br>$f(x,y) = \begin{cases} c, & \text{if } x^2 + y^2 \leq R^2 \\ 0, & \text{if } x^2 + y^2 > R^2 \end{cases}$ | 5  | CO3 |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                          |    |     |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----|
|     |  <p>for some value of <math>c</math>.</p> <p>(i) Compute the probability that <math>D</math>, the distance from the origin of the point selected, is less than or equal to <math>a</math>.</p> <p>(ii) Find <math>E[D]</math>.</p>                                                                                                      |    |     |
| 3b  | Let $Y = X^2$ . Find and sketch the pdf of $Y$ if $X$ is a uniform random variable over $(-1, 2)$ .                                                                                                                                                                                                                                                                                                                      | 5  | CO3 |
| 3c  | Let $X$ and $Y$ be independent uniform random variables over $(0,1)$ . Find the pdf of $Z=XY$ .                                                                                                                                                                                                                                                                                                                          | 5  | CO3 |
| Q 4 | Attempt any 2 parts of the following                                                                                                                                                                                                                                                                                                                                                                                     | 10 |     |
| 4a  | <p>(i) Write a short note on Wiener Process. State the expectation, variance, autocorrelation and auto-covariance function for a Wiener Process <math>X(t)</math>.</p> <p>(ii) Differentiate between Strict Sense Stationary and Wide Sense Stationary Process.</p>                                                                                                                                                      | 5  | CO4 |
| 4b  | <p>(i) Consider the random process</p> $X(t) = A \cos(\omega t + \theta), \quad -\infty < t < \infty$ <p>Where <math>A</math> and <math>\omega</math> are constants and <math>\theta</math> is a uniform random variable over <math>(-\pi, \pi)</math>. Is <math>X(t)</math> a Wide Sense Stationary process or not?</p> <p>(ii) Define Birth-Death Process with state transition diagram and equilibrium equations.</p> | 5  | CO4 |
| 4c  | <p>Patients arrive at the doctor's office according to a Poisson process with rate <math>\lambda = 1/10</math> minute. The doctor will not see a patient until at least 3 patients are in the waiting room.</p> <p>(i) Find the expected waiting time until the first patient is admitted to see the doctor.</p> <p>(ii) What is the probability that nobody is admitted to see the doctor in the first hour?</p>        | 5  | CO4 |
| Q 5 | Attempt any 2 parts of the following                                                                                                                                                                                                                                                                                                                                                                                     | 10 |     |

3

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |     |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|
| 5a | <p>Consider the Markov chain shown below:</p>  <p>(i) Find <math>P(X_4=3   X_3=2)</math></p> <p>(ii) Find <math>P(X_3=1   X_2=1)</math></p> <p>(iii) If we know <math>P(X_0=1)=1/3</math>, find <math>P(X_0=1, X_1=2)</math></p> <p>(iv) If we know <math>P(X_0=1)=1/3</math>, find <math>P(X_0=1, X_1=2, X_2=3)</math>.</p>                                                                                                                                                                                                                                                                             | 5 | CO5 |
| 5b | <p>(i) A drive-in banking service is modelled as an M/M/1 queueing system, with customer arrival rate of 2 per minute. It is desired to have fewer than 5 customers line up 99 % of the time. How should the service rate be?</p> <p>(ii) A corporate computing center has 2 computers of the same capacity. The jobs arriving at the center are of two types: internal jobs and external jobs. These jobs have Poisson arrival times with rates 18 and 15 per hour respectively. The service time for a job is an exponential r.v. with mean 3 minutes. Find the average waiting time per job when one computer is used exclusively for internal jobs and the other for external jobs.</p> | 5 | CO5 |
| 5c | <p>Customers arrive at a watch repair shop according to a Poisson process at a rate of 1 per every 10 minutes, and the service time is an exponential r.v. with mean 8 minutes.</p> <p>(i) Find the average number of customers <math>L</math>, the average time a customer spends in the shop <math>W</math>, and the average time a customer spends in waiting for service <math>W_q</math>.</p> <p>(ii) Suppose that the arrival rate of the customers increases 10 percent. Find the corresponding changes in <math>L</math>, <math>W</math>, and <math>W_q</math>.</p>                                                                                                                 | 5 | CO5 |

4